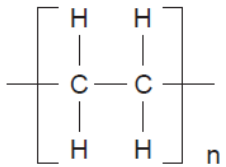
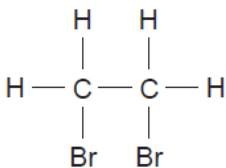
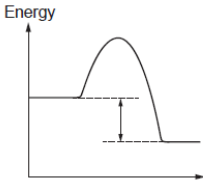
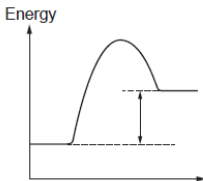
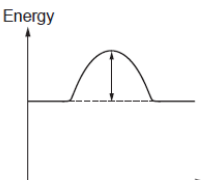


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		C_nH_{2n}	1			1		
		(ii)	I		1			1		
			II		1			1		
			III	addition neutral answer – additional	1			1		
		(iii)		$C_2H_4 + 3O_2 \longrightarrow \boxed{2} CO_2 + \boxed{2} H_2O$ award (1) for product award (1) for balancing only if product correct		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(b)	(i)		$612 + (4 \times 412) + 436 (= 2696)$ (2) if answer incorrect award (1) for clear indication of the number of each type of bond e.g. $C=C + [4 \times (C-H)] + H-H$ award (1) when one error made in identifying the number of each type of bond e.g. $612 + 412 + 436 / 1460$ $612 + (4 \times 412) / 2260$		2		2	2	
		(ii)		348 (2) if answer incorrect award (1) for either of following 6×412 2472		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(b)	(iii)		 ☒						
				 ☐						
				 ☐						
	(c)			but-1-ene C but-2-ene A 2-methylpropene B award (2) for all three correct award (1) for any one correct	2			2		
				Question 7 total	6	7	0	13	4	0